Swad Mustasin Khan

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RESEARCH INTERESTS

Machine learning and security: securing LLM-based agents (prompt injection, memory poisoning, tool/MCP misuse); robust and privacy-preserving federated learning for critical-infrastructure intrusion detection; calibrated, reproducible security measurement.

EDUCATION

M.Sc. Applied Cybersecurity, Erasmus Mundus Joint Master (CyberMACS) Sep 2025 – Oct 2027 (expected)
Kadir Has University, Istanbul, Türkiye (Year 1) · GPA **3.32 / 4.00**
Ss. Cyril and Methodius University in Skopje, North Macedonia (Year 2) · Web Security and Cryptology track
Selected courses: Applied Cryptography, System Exploitation and Penetration Testing, Cyber Warfare, Research Methods, Computer Networks and Security.

B.Sc. Computer Science and Engineering Jan 2019 – Apr 2023
BRAC University, Dhaka, Bangladesh · CGPA **3.54 / 4.00** · 136 credits
Thesis: *SAAD: An Approach to Detect Smartphone Addiction through Activity Recognition and App Usage Behaviour*

PUBLICATIONS

[1] S. M. Khan, S. Labastie, and O. Ceylan, "Meta-heuristic optimization approach for cyber-physical defense in smart grids," presented at the *1st Int. Conf. Cybersecurity, Data Science, and Machine Learning (ICCDM-2026)*, Universiti Putra Malaysia, Apr. 23–24, 2026. Springer, in press.

[2] S. M. Khan *et al.*, "SAAD: An approach to detect smartphone addiction through activity recognition and app usage behaviour," under review, *Frontiers*.

[3] S. M. Khan and O. Ceylan, "Benchmarking secure aggregation in federated learning: A calibrated measurement harness for the communication, latency and privacy trade-off on a physical edge testbed," manuscript in preparation.

RESEARCH EXPERIENCE

Research Assistant, TÜBİTAK 1001 project Dec 2025 – Sep 2026
"Multi-objective Encryption Optimization in Secure Federated Learning for Critical Infrastructures" · Kadir Has University · Supervisor: Assoc. Prof. Oğuzhan Ceylan

- Built a measurement harness for secure aggregation in federated intrusion detection (Edge-IIoT) on a four-node Raspberry Pi testbed, with byte-level communication accounting and a simulator validated against hardware.
- Derived a testbed noise floor from a known-zero invariant (secure aggregation minus baseline) and used it to calibrate every comparison; replication with it overturned a single-run headline result.
- Showed additive masking costs +4.8% uplink (metadata only) and under 0.34 ms per round, and that an 8-round schedule matches 30-round detection quality with 3.7× less uplink (TPE and NSGA-II search via Optuna).
- Evaluated update-level differential privacy with an RDP accountant validated against Opacus; showed it is infeasible at three clients.

Research case study: Cyber-physical defence in smart grids Kadir Has University
With Assoc. Prof. Oğuzhan Ceylan · led to publication [1]

- Applied meta-heuristic algorithms (GA, PSO) to mitigate cyber-physical attacks on OpenDSS distribution-grid simulations, compared with threshold-based detection.

B.Sc. thesis research (SAAD) Mar 2022 – Apr 2023

- Collected smartphone usage and sensor data from 30 participants over 7 days; combined K-Means/K-Medoids clustering with LSTM activity recognition, validated against SAS-SV scores; 28% of participants showed addictive behaviour.

SELECTED PROJECTS

MCP-Guard: static security scanner for MCP servers 2026 · github.com/SwadKhan/MCP_Guard

- Five rules mapped to the OWASP Top 10 for LLM Applications 2025: tool poisoning, excessive permissions, rug-pull detection (SHA-256 lockfiles), unsanitised tool output, hard-coded secrets. CLI with CI gating; 56 tests; MIT licence.

Eco-Agent: LLM-based civic air-quality platform Hacktoberfest 2026, Skopje

- Live sensor data, anonymous pollution reports and AI-written municipal briefs; anonymous by design (no accounts, no IP logging). LangChain, open-weight LLM via Groq.

ShipOfflineAuth: offline credential system IACyC 2026 hackathon

- Team project (Sinking Ship) for offline ship credentialing via printed QR codes; received the Best Collaborators award.

ChatBotPRO: multi-modal retrieval-augmented generation 2025 · github.com/SwadKhan/ChatBotPRO

- RAG over PDFs, images (OCR) and slides with citations. LangChain, ChromaDB, FastEmbed, LLaMA 3.1 8B, Streamlit.

HONOURS AND AWARDS

- Erasmus Mundus Joint Master scholarship (European Union), CyberMACS, 2025–2027.
- Best Collaborators award, IACyC 2026 hackathon, Kadir Has University, 2026.
- BRAC University merit scholarship (50%), 10 consecutive semesters, 2020–2023; Vice-Chancellor's List (2 semesters) and Dean's List (2 semesters).
- 2nd Runner-up, MIST Inter-University ICT Innovation Fest, 2021.

TEACHING AND INDUSTRY

Undergraduate Student Tutor, BRAC University 2022 – 2024

Programming Language II (CSE111), Data Structures (CSE220), Algorithms (CSE221), Mathematics I (MAT110).

Software Engineer Intern, BRAC Bank PLC, Dhaka Jul 2023 – Oct 2023

Built web application prototypes (Java, JavaScript) with senior engineers and drafted system configuration specifications.

SKILLS AND CERTIFICATIONS

ML / privacy	Python, scikit-learn, PyTorch, Optuna, federated learning, differential privacy (RDP accounting)
LLMs / agents	LangChain, RAG (ChromaDB), Model Context Protocol, OWASP Top 10 for LLM Applications
Security	Kali Linux, Burp Suite, Nmap, Metasploit, Wireshark; Cisco CCNA (2026); TryHackMe Jr Penetration Tester and Web Application Pentesting paths; HTB Academy AI Red Teamer path (in progress)
Systems / tools	Linux, Raspberry Pi, Tailscale, OpenDSS, Git, LaTeX; Java, JavaScript (MERN), MySQL
Languages	Bengali (native); English: IELTS Academic 7.5, CEFR C1 (2024)

REFEREES

Assoc. Prof. Oğuzhan Ceylan, Head of Dept. of Management Information Systems, Kadir Has University (oguzhan.ceylan@khas.edu.tr) · Prof. Hasan Dağ, Kadir Has University